

Lifting Manual

A visual reference for rigging hardware and crane lifts

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Rigging hardware

Crane lifts

Key numbers

Sling angles

1. What It Is

The Lifting Manual is a **visual reference**, not a form. It condenses the main rules from two standards into note cards with icons, colour chips, tables and a sling-angle drawing. It is reached from Safety › Process Safety › Lifting Operations › Lifting Manual and has two tabs.

Tab	Based on
Rigging Hardware	The company's rigging-hardware standard (GI 7.029), with ASME B30.9 for slings
Crane Lifts	The company's crane-lift standard (GI 7.028) and OISD-RP-205

Each tab opens with a **key numbers** card that can be folded away, then groups of rules.

2. Rigging Hardware Tab

Key numbers to remember

- **Minimum design safety factor 5:1** for slings and hardware.
- **Proof-load test:** the maker tests every sling at double its rated load.
- **Periodic inspection:** a certified rigger inspects and records each sling at least every six months.
- **Minimum sling angle 30°:** the legs of a sling must never be flatter, or the legs are overloaded.

What else is covered

Topic	Highlights
Sling angle rule	A drawing shows how the load on each leg grows as the angle gets flatter

Topic	Highlights
Sling identification tags	What each sling type's tag must show: chain (maker, grade, reach, legs, working load and angle), wire rope (maker, working load) and synthetic or web (maker, date, material, ratings for each hitch)
Wire rope rules	Only Flemish-eye with swaged ends; no fold-back eyes, no knotted eyes, no aluminium ferrules
Chain temperature limits	The temperature range outside which alloy chain must not be used
Inspection levels	Frequent: before every use by a certified rigger, no record needed. Periodic: every six months at most, recorded on the sling inspection log
Strict prohibitions	Repaired slings, knotted shortening, re-splicing, wire rope clips, in-house fasteners, aluminium ferrules, fold-back eyes
Storage, use and handling	Self-closing latches on all hooks designed for them; slings not shortened except by approved methods; tags kept legible
Standards reference	ASME B30.9 as the minimum; proof-load certificates kept for the sling's life; chain made to the relevant chain standard

3. Crane Lifts Tab

Key numbers to remember

Number	Rule
40 tonnes	A load of 40 tonnes or more needs a full Critical Lift Plan , with no exceptions
85%	A load above 85% of the crane's rated capacity is also a critical lift
Tandem de-rate 25%	When two or more cranes lift together, reduce each crane's allowed load by 25%
Manbasket de-rate 50%	When lifting a basket with people, the crane may use only half its capacity

When a lift counts as critical

Examples listed include working within 10 metres of hydrocarbon or pressurised piping, populated or traffic areas, or a railway; any boom attachment fitted to the main boom; a load of 40 tonnes or more; a load above 85% of rated capacity; explosion, fire or high-heat hazards; and working near energised power lines. A Rigger-I must be in charge of coordination.

Other topics

Topic	Highlights
Power-line approach distances	Side-by-side tables for the two standards. Minimum clearances rise with voltage, from about 3 metres (10 feet) up to 7.6 metres (25 feet) in the company standard
Rigger certification levels	Three levels, valid up to two years. Level III may rig up to 10 tonnes without supervision; Level II up to 40 tonnes and may originate critical plans; Level I approves critical plans and supervises them
Tandem lifts	De-rate all cranes by 25%, use cranes of the same size, make and model where possible, a separate approved plan each time, and a Rigger-I in charge
Man-basket rules	De-rate by 50%, a completed permit on site, only when other access is more hazardous. Maximum four persons, test load twice the working load, weekly inspection, life jackets, and night transfer in emergencies only
Mandatory crane equipment	Fire extinguisher, back-up alarm, spark arrester where needed, rated capacity limiter, anti-two-block device, aircraft warning light above 60 metres, load charts in the cab and knot-free tag lines
Operator qualification	Valid certificate, driving licence, training, fitness, knowledge of signals. Never leave the controls with a load attached
Ground and trenches	Keep a distance from a trench equal to its depth in firm soil, and twice its depth in softer soils
Golden rules	Trial lift before every lift, no mobile phones, one signaller only, no welding on hooks, mats under all pads, nobody rides the hook or load
Wind speed limits	Different limits for different operations
PPE	Hard hat, gloves, glasses, overalls, safety shoes, a harness for work above about 3 metres, and life vests for basket transfers

A summary, not the standard

The manual is a quick-reference summary. The full company and industry standards, and site procedures, always take precedence.

4. Design

- A fixed header and a tab bar, with a button to fold or unfold every card.
- Colour chips and icons make the “do” and “do not” items easy to separate.
- The sling-angle diagram is drawn by the app: boxes of fixed width with anchors at different heights show how the angle changes, and the colour moves from safe to caution to danger as the angle gets flatter.